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| CONTACT INFORMATION | <div><div><div>✉ E-mail: <a href="mailto:doserjef@msu.edu">doserjef@msu.edu</a></div><div>Tel: (585) 683-4170</div></div><div><div>Homepage: <a href="https://www.jeffdoser.com">https://www.jeffdoser.com</a></div><div>GitHub: <a href="https://github.com/doserjef">https://github.com/doserjef</a></div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| RESEARCH INTERESTS  | Statistical Ecology, Wildlife and Natural Resource Conservation, Bayesian Modeling, Biodiversity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| EDUCATION           | <div><div>Michigan State University, East Lansing, MI</div><div><div><div>• Ph.D., Forestry and Ecology, Evolution, and Behavior</div><div>Completed 2022</div><div><div>• Dissertation title: Development and application of hierarchical models for monitoring avian soundscapes, populations, and communities</div><div>• Advisor: Andrew O. Finley</div></div></div><div>• M.S., Applied Statistics</div><div>Completed 2021</div></div><div><div>State University of New York at Geneseo, Geneseo, NY.</div><div><div>• B.S. Mathematics and Biology</div><div>Summa Cum Laude</div><div>Completed 2018</div></div></div></div>                                                                                                                                                                                                                                             |
| RESEARCH EXPERIENCE | <div><div>Postdoctoral Research Associate, Michigan State University</div><div>2022 - Current</div><div><div>• Department of Integrative Biology</div><div>• Advisor: Elise F. Zipkin</div></div><div><div>Postdoctoral Fellow: Institute of Biodiversity, Ecology, Evolution, and Macrosystems,</div><div>Michigan State University</div><div>2022 - 2023</div></div><div><div>Graduate Research Assistant, Michigan State University</div><div>2018-2022</div><div><div>• Department of Forestry</div><div>• Advisor: Andrew O. Finley</div></div><div><div>Undergraduate Research Assistant, SUNY Geneseo</div><div>2015-2018</div><div>• Advisors: Kristina M. Hannam, David D. Meisel</div></div></div></div>                                                                                                                                                               |
| TEACHING EXPERIENCE | <div><div>Guest Lecturer, Michigan State University</div><div><div>• IBIO 860: Modern Statistical Models in Ecology</div><div>Feb 2023</div><div>• Two guest lectures on capture-recapture models</div></div><div><div>Co-Instructor, Michigan State University</div><div><div>• IBIO 831 Statistical Methods in Ecology and Evolution II</div><div>2022</div><div>• FOR/STT 875 R Programming for Data Sciences</div><div>2020-2022</div></div><div><div>Teaching Assistant, Michigan State University</div><div><div>• FOR/STT 875 R Programming for Data Sciences</div><div>2018-2020</div><div>– Lead author of <a href="#">course textbook</a>.</div><div>• FOR 472 Ecological Monitoring and Data Analysis</div><div>2018</div></div><div><div>Undergraduate Lab Assistant, SUNY Geneseo</div><div>2016</div><div>• Introductory Biology Lab</div></div></div></div></div> |

UNDER  
REVIEW

14. Zipkin, E.F., **Doser, J.W.**, Davis, C.L., Leuenberger, W.L., Ayebare, S., Davis, K.L. Integrated community models: A framework combining multi-species data sources to estimate the status, trends, and dynamics of biodiversity. *Journal of Animal Ecology*.
13. Ayebare, S., **Doser, J.W.**, Plumptre, A., Owunji, I., Mugabe, H., Zipkin, E.F. An environmental habitat gradient and within-habitat segregation enable co-existence among ecologically similar bird species. *Proceedings of Royal Society B*.
12. Roberts, C.P., **Doser, J.W.**, Berry, L.L., Fowler, A., Marshall, P.M., Middaugh, C., Rowe, K., Schmit, J., Shaw, M., Wilson, K. Scenario planning and multi-species occupancy models reveal positive avian responses to restoration of afforested woodlands. *Restoration Ecology*.
11. **Doser, J.W.**, Kéry, M., Finley, A.O., Saunders, S.P., Weed, A.W., Zipkin, E.F. Guidelines for the use of spatially-varying coefficients in species distribution models. *Global Ecology and Biogeography*. Preprint: [arXiv:2301.05645](https://arxiv.org/abs/2301.05645).
10. **Doser, J.W.**, Finley, A.O., Banerjee, S. Joint species distribution models with imperfect detection for high-dimensional spatial data. *Ecology*. Preprint: [arXiv:2204.02707](https://arxiv.org/abs/2204.02707).
9. Williamson, C.R., Campa III, H., **Doser, J.W.**, Winterstein S.R., Beyer Jr, D.E. Elk space-use and resource selection patterns in response to summer trail-based recreation. *Journal of Wildlife Management*.

PEER  
REVIEWED  
PUBLICATIONS

8. **Doser, J.W.**, Finley, A.O., Kéry, M., Zipkin, E.F (2022). **sp0ccupancy**: An R package for single-species, multi-species, and integrated spatial occupancy models. *Methods in Ecology and Evolution*. 13(8), 11670-1678. <https://doi.org/10.1111/2041-210X.13897>.
7. **Doser, J.W.**, Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F (2022). Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. *Methods in Ecology and Evolution*. 13(4), 919-932. <https://doi.org/10.1111/2041-210X.13811>.
6. **Doser, J.W.**, Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. (2021). Trends in bird abundance differ among protected forests but not bird guilds. *Ecological Applications* 31(6):e02377. <https://esajournals.onlinelibrary.wiley.com/doi/abs/10.1002/eap.2377>.
5. **Doser, J. W.**, Finley, A. O., Weed, A. S., Zipkin, E. F. (2021). Integrating automated acoustic vocalization data and point count surveys for estimation of bird abundance. *Methods in Ecology and Evolution*, 12(6), 1040-1049. <https://doi.org/10.1111/2041-210X.13578>.
4. Hoffmann, H.M., Meadows, J.D., Breuer, J.A., Yaw, A.M., Nguyen, D., Tonsfeldt, K.J., Chin, A.Y., Devries, B.D., Khan, R., Trang, C., Oosterhouse, H.J., Sora Lee, J., **Doser, J.W.**, Gorman, M. R., Welsh, D.K., Mellon, P. L. (2021). The transcription factors VAX1 and SIX3 are required for suprachiasmatic nucleus circadian output and fertility in female mice. *Journal of Neuroscience Research*. <https://doi.org/10.1002/jnr.24864>.
3. Teimouri, M., **Doser, J.W.**, Finley A.O. (2020) **ForestFit**: An R package for modeling plant size distributions. *Environmental Modelling & Software*, 131, 104668. <https://doi.org/10.1016/j.envsoft.2020.104668>.
2. **Doser, J.W.**, Finley, A.O., Kasten, E.P., Gage, S.H. (2020). Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. *Ecological Indicators*, 113, 106244. <https://doi.org/10.1016/j.ecolind.2020.106244>.
1. **Doser, J.W.**, Hannam, K.M., Finley, A.O., (2020). Characterizing functional relationships between anthropogenic and biological sounds: A western New York state soundscape case study. *Landscape Ecology*, 35(3), 689-707. <https://doi.org/10.1007/s10980-020-00973-2>.

IN  
PREPARATION

- **Doser, J.W.**, Saunders, S.P., Finley, A.O., Bateman, B.L., Grand, J., Weed, A.S., Zipkin, E.F. Nonstationary effects of climate and land use on forest bird communities in the eastern United States.
- **Doser, J.W.**, Finley, A.O., Kéry, M., Zipkin, E.F. **spAbundance**: An R package for single-species and multi-species spatially-explicit abundance models.
- Finley, A.O., **Doser, J.W.**, Itter, M.S., Stanke, H., Domke, G.M. High-dimensional spatial models for predicting species-specific forest biomass across the continental United States.

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| BOOKS UNDER CONTRACT    | Finley, A.O. and <b>Doser, J.W.</b> <i>Introduction to Forestry Data Analysis with R</i> . Chapman & Hall CRC. Expected completion date: March 2023. <b>Draft available upon request.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| PUBLISHED SOFTWARE      | <b>spOccupancy</b> . <b>Doser, J.W.</b> , Finley, A.O. (Nov 2021 - current). Single-species, Multi-species, and Integrated Spatial Occupancy Models. Downloaded 7045 times as of March 3, 2023. <a href="https://cran.r-project.org/web/packages/spOccupancy/index.html">https://cran.r-project.org/web/packages/spOccupancy/index.html</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| PROFESSIONAL POSITIONS  | <p><i>Statistical Consultant</i>, Michigan State University 2019-2020</p> <ul style="list-style-type: none"> <li>• College of Agriculture and Natural Resources Statistical Consulting Center</li> <li>• Provided statistical and programming guidance to undergraduate students, graduate students, and faculty members.</li> </ul> <p><i>Student Database Programmer</i>, SUNY Geneseo 2016-2018</p> <ul style="list-style-type: none"> <li>• Computing and Information Technologies Department</li> <li>• Developed, modified, and tested Banner applications using SQL, Groovy, SQR, and the Argos Enterprise Reporting System</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| RELEVANT SKILLS         | <p><i>Programming Languages/Statistical Software</i>: R, C++, C, JAGS, NIMBLE</p> <p><i>Familiar Languages</i>: SQL, SAS</p> <p><i>Software</i>: Latex/Knitr, R Markdown/bookdown, R Shiny</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| GRANTS AND SCHOLARSHIPS | <p><i>Submitted</i>: USDA Agriculture and Food Research Initiative: <i>A national-scale early detection and ecosystem service impact assessment tool for invasive plants</i>. Statistical consultant.</p> <p><i>Funded</i>: NSF, DEB, Collaborative Research MRA: <i>Estimating and forecasting nonstationary, multi-scale climate and land-use effects on avian communities</i>. Postdoctoral Researcher. 2022-2027. (\$1,306,419)</p> <p><i>Funded</i>: US National Park Service. <i>Developing spatially explicit integrated models to evaluate avian distributions and regional bird population dynamics in eastern US National Parks</i>. Postdoctoral Researcher/ Graduate Student Mentor. 2022-2026. (\$228,084)</p> <p><i>Not selected</i>: Northeast Climate Adaptation Science Center, USGS. <i>Improving bird and bat conservation in forest resilience planning: A Northeast climate change adaptation strategy</i>. Postdoctoral Researcher.</p> <p>Barry M. Goldwater Scholarship. 2017-2018. (\$7,500)</p> <p>Dr. Wendell and Barbara Rhodes Research Award. 2016. (\$8,150)</p> <p>Geneseo Foundation Scholarship. 2014-2018. (\$65,000)</p> |  |
| AWARDS AND HONORS       | <p>David R. Anderson Outstanding Student Paper Award, Biometrics Working Group, The Wildlife Society 2022</p> <p>SUNY Chancellor's Award for Student Excellence 2018</p> <p>Excellence in Mathematics, SUNY Geneseo 2018</p> <p>Dr. Sam Molnar Scholar Athlete Award, SUNYAC 2018</p> <p>James Fulton Award for Academic and Athletic Excellence, SUNYAC 2018</p> <p>Phi Beta Kappa Honors Society, SUNY Geneseo 2017</p> <p>SUNY Chancellor's Scholar-Athlete Award 2016-2018</p> <p>Edgar Fellows Honors Program 2014-2018</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |

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| UNIVERSITY<br>SERVICE                          | Ecology, Evolution, and Behavior Program Research Symposium Chair, Michigan State University,<br>2021-2022                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                | Ecology, Evolution, and Behavior Program Graduate/Postdoc Awards Committee, Michigan State<br>University, 2020-2022                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                | Ecology, Evolution, and Behavior Program Research Symposium Committee, Michigan State Uni-<br>versity 2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                | Treasurer, Graduate Student Organization, Department of Forestry, Michigan State University 2019-<br>2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                | Student Athlete Advisory Committee, SUNY Geneseo 2015-2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| EDITORIAL<br>SERVICE                           | <b>Peer Reviews:</b> <i>Nature Ecology and Evolution</i> (1), <i>Methods in Ecology and Evolution</i> (4), <i>Ecological Monographs</i> (2), <i>Ecological Applications</i> (1), <i>Ecography</i> (2), <i>Ecology</i> (1), <i>Conservation Biology</i> (1), <i>Landscape Ecology</i> (1), <i>Biological Invasions</i> (1), <i>Diversity and Distribution</i> (1), <i>Ecological Indicators</i> (2), <i>Animal Conservation</i> (2), <i>Ecology and Evolution</i> (1), <i>Ornithological Applications</i> (1), <i>Biodiversity and Conservation</i> (1), <i>Journal of Ornithology</i> (1), <i>PLOS ONE</i> (1), <i>Ekológia</i> (1) |
| INVITED<br>WORKSHOPS<br>AND PRESEN-<br>TATIONS | Finley, A. O., <b>Doser, J. W.</b> May 2023. Bayesian modeling for environmental spatial data analysis. Conference on Applied Statistics and Natural Resources. West Lafayette, Indiana. Workshop.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                | <b>Doser, J. W.</b> , Finley, A. O., Zipkin, E. F. March 2023. High-dimensional spatial models for predicting forest biomass and bird distributions across the continental US. Levi, Finland. BayesComp 2023. Oral presentation.                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                | <b>Doser, J. W.</b> January 2023. Hierarchical models for high-dimensional ecological data across large spatial domains. Bucknell University Department of Mathematics Distinguished Visiting Professor Program. Lewisburg, PA. Seminar.                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                | <b>Doser, J. W.</b> November 2022. Using spatially-explicit occupancy models to understand occurrence and biodiversity patterns across macroscales. University of Arkansas Biology Seminar Series. Fayetteville, Arkansas. Seminar.                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                | <b>Doser, J. W.</b> , Finley, A. O. October 2022. Spatially-explicit occupancy modeling with the <b>sp0ccupancy</b> R package. Ecological Society of America Statistical Ecology Webinar Series. Remote. Workshop.                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                | <b>Doser, J. W.</b> , Finley, A.O., Banerjee, S. August 2022. Joint species distribution models with imperfect detection for high-dimensional spatial data. Joint Statistical Meetings. Washington, D. C. Oral presentation.                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                | <b>Doser, J. W.</b> July 2022. Bayesian occupancy modeling with the <b>sp0ccupancy</b> R package. Living Landscapes Lab, University of Arkansas. Remote. Workshop.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                | <b>Doser, J. W.</b> July 2022. Bayesian occupancy modeling with the <b>sp0ccupancy</b> R package. Tel Aviv University. Tel Aviv, Israel. Workshop.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                | <b>Doser, J. W.</b> June 2022. Bayesian occupancy modeling with acoustic data in <b>sp0ccupancy</b> . Cornell Acoustic Methods Working Group Series. Remote. Workshop.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                | <b>Doser, J. W.</b> , Finley, A. O., Zipkin, E.F. November 2023. Spatially-explicit occupancy modeling with the <b>spOccupancy</b> R package. The Wildlife Society Annual Conference. Louisville, Kentucky. Workshop.                                                                                                                                                                                                                                                                                                                                                                                                               |
| OTHER<br>WORKSHOPS<br>AND PRESEN-<br>TATIONS   | <b>Doser, J. W.</b> , Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F. August 2022. Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. Ecological Society of America. Montréal, Canada. Oral presentation.                                                                                                                                                                                                                                                                                                                                 |
|                                                | <b>Doser, J. W.</b> , Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F. June 2022. Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. International Statistical Ecology Conference. Remote. Oral presentation.                                                                                                                                                                                                                                                                                                                              |

- Doser, J.W.**, Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. July 2021. Trends in bird abundance differ among protected forests but not bird guilds. American Ornithological Society Meeting. Remote. Oral presentation.
- Doser, J.W.**, Finley, A.O., Kasten, E. P., Gage, S. H. June 2021. Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. International Ecoacoustics Congress. Remote. Oral presentation.
- Doser, J.W.**, Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. May 2021. Trends in bird abundance differ among protected forests but not bird guilds. Ecology, Evolution, and Behavior Program Symposium. Michigan State University. Remote. ***Awarded 2nd Place for Best Oral Presentation.***
- Doser, J.W.** February 2021. Forest bird monitoring using acoustic recordings and hierarchical models. Hanover Seminar Series. Department of Forestry. Michigan State University. Oral presentation.
- Doser, J.W.**, Finley, A.O., Kasten, E. P., Gage, S. H. February 2020. Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. Michigan State University. Oral presentation.
- Doser, J.W.**, Hannam, K.M., Finley, A.O. September 2019. Using hierarchical Bayesian models to analyze acoustic data. XXVII International Bioacoustics Congress, Brighton, England. Oral presentation.
- Doser, J.W.**, Hannam, K.M., Finley, A.O. May 2019. Characterizing functional relationships between biophony and technophony: a western New York soundscape case study. Northeast Regional Environmental Acoustics Symposium. Providence, RI. Oral presentation.
- Doser, J.W.**, Meisel, D.D. April 2018. Developing an automated bird song recognition system using wavelets and simple machine learning techniques. Geneseo Recognizing Excellence, Achievement and Talent Day, Geneseo, NY. Oral presentation.
- Doser, J.W.**, Hannam, K.M. October 2017. Using soundscape maps to assess a local soundscape. Middle States Division American Association of Geographers Annual Meeting, Geneseo, NY. Poster presentation.
- Doser, J.W.**, Hannam, K.M. April 2017. Analysis of seasonal temporal variation of soundscapes in western New York. Northeast Natural History Conference, Cromwell, CT. Oral presentation.
- Doser, J.W.**, Hannam, K.M. August 2016. Use of soundscape recordings to analyze daily temporal variation in avian vocalizations in western New York. North American Ornithological Conference, Washington D.C. Poster presentation.

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| OUTREACH     | Volunteer, Delta 4H Wildlife Club, East Lansing, MI                                                                                           | Spring 2019  |
|              | Guest Speaker, French Road Elementary School, Brighton, NY                                                                                    | 2018         |
|              | <ul style="list-style-type: none"> <li>• Discussed applications of Raspberry Pi computers and how to use them for sound recording.</li> </ul> |              |
| AFFILIATIONS | The Wildlife Society                                                                                                                          | 2022-current |
|              | American Statistical Association                                                                                                              | 2019-current |
|              | Ecological Society of America                                                                                                                 | 2021-current |
|              | American Ornithological Society                                                                                                               | 2021-current |
|              | International Society of Ecoacoustics                                                                                                         | 2019-2022    |
| REFERENCES   | <b>Andrew O. Finley</b><br>Michigan State University<br>E-mail: finleya@msu.edu                                                               |              |
|              | <b>Elise F. Zipkin</b><br>Michigan State University<br>E-mail: ezipkin@msu.edu                                                                |              |

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